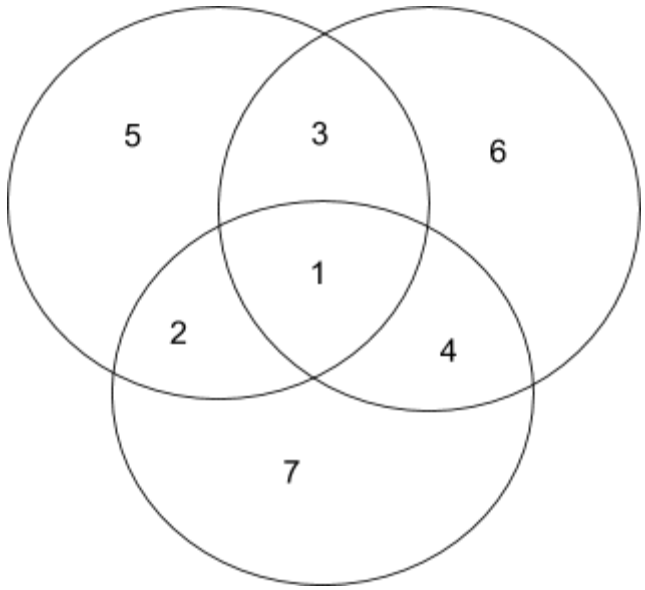
CS230-HW08 **<u>Due date: November 16</u>**

Dr. Lutz and Slutzki

- Read Lecture Notes Chapter 5 on Relations.
- Watch the videos pertaining to Chapter 5 of Lecture Notes.

(01) This problem concerns the (7,4) Hamming code discussed in the lectures. Using the diagram



to encode and decode strings, show your work clearly in the following.

a) If the message is 1001, what codeword should be sent?

b) If the string 0001110 is received and at most one bit was flipped, what was the original message?

(02) Recall from the lectures that $\{0,1\}^n$ is the set of all n-bit binary strings and that $d_H(x,y)$ is the Hamming distance between to string $x,y \in \{0,1\}^n$. Let $A \subseteq \{0,1\}^9$, with $|A| = 12$. Prove that there exists $x,y \in A$ such that $1 \leq d_H(x,y) \leq 4$.

(03) Let P be some non-empty population of people. Define binary relations on the set P :

- $M = \{(a,b) \mid a = b \text{ or } a \text{ is a mother of } b\}$,
- $S = \{(a,b) \mid a \text{ and } b \text{ are siblings}\}$, [siblings have the same mother]

- $R = \{(a,b) \mid a \neq b\}$.
- $F = \{(a,f) \mid f$ is the father of $a\}$ [every person has a unique father]
- $N = \{(a,b) \mid a$ and b have the same first name$\}$
- Determine the following properties of the three relations (write in the symbol "#" in appropriate entries of the given table).

	M	*S*	*R*	*F*	*N*
reflexive					
symmetric					
asymmetric					
antisymmetric					
transitive					
equivalence relation					

(04) Let $E = \{(1,1),(3,2)(3,5),(4,3),(5,5),(6,2)\}$ be a binary relation on the set $\{1,2,3,4,5,6\}$.

a) Draw a graph representation of *E*.
b) Write a Boolean matrix representation of *E*.
c) Compute the smallest reflexive relation that contains *E*. Give a graph and the matrix.
d) Draw the smallest symmetric relation that contains *E*. Give a graph and the matrix.
e) Draw the smallest transitive relation that contains *E*. Give a graph and the matrix.
f) Draw the smallest equivalence relation that contains *E*. Give a graph and the matrix.

(05) Consider the set $X = \{2,3,4,5,6,7,8\}$ and define the following binary relation $R \subseteq X \times X$:

$(x,y) \in R \Leftrightarrow 3 \mid x - y$.

a) List *R* as a set of ordered pairs.
b) Prove that *R* is an equivalence relation.
c) Draw the graph representation of *R* .

(06) Let A be a non-empty set and let $R, S \subseteq A \times A$ be binary relations on *A*. Define new relations $T, T' \subseteq A \times A$ by $T = R \cap S$ and $T' = R \cup S$.

<u>Prove</u> or <u>Disprove</u>: if R and S are equivalence relations, then so are T and T' .

(07) Consider the set $S = \{0,1,2,3,4\}$.

a) Construct the equivalence relation corresponding to the partition $\Pi = \{\{0,2\},\{1\},\{3,4\}\}$ and represent it as a set of ordered pairs.

b) Construct the equivalence relation corresponding to the partition $\Pi = \{\{0\},\{1,2,3,4\}\}$ and represent it as a directed graph.

c) Construct the equivalence relation corresponding to the partition $\Pi = \{\{0\},\{2\},\{1,3,4\}\}$ and represent it as a set of ordered pairs and as a directed graph.

(08) Let $S = \{1,2,...,n\}$ and $\mathbf{A} = \mathsf{P}(S) - \{\varnothing\}$. In words, the set $\mathbf{A}$ consists of all non-empty subsets of *S*. For each set $A \in \mathbf{A}$, let $m(A)$ to be the smallest element of the set *A*.

a) Define a binary relation $R \subseteq \mathbf{A} \times \mathbf{A}$ such that $(A,B) \in R \Leftrightarrow m(A) = m(B)$. Prove that *R* is an equivalence relation on $\mathbf{A}$.

b) What is the cardinality of the quotient $\mathbf{A}/R$?

c) Define a binary relation ρ on $\mathbf{A}$ by: $(A,B) \in \rho \Leftrightarrow [A = B \vee m(A) < m(B)]$. Prove that ρ is a partial order on *A*. **Hint**. Show that ρ is reflexive, antisymmetric and transitive.

d) Is ρ a total order? Why?